

How the transitioning to sustainable food system affects zoonoses:

Highly Pathogenic Avian influenza and outdoor poultry farming

1) Background

- Highly pathogenic avian influenza (HPAI) is a zoonosis that can infect both birds and human.
- Changes towards more extensive poultry farming may affect HPAI outbreaks
- The changes include **outdoor farming, farm size reduction and decrease in number of farm.**

2) Methods

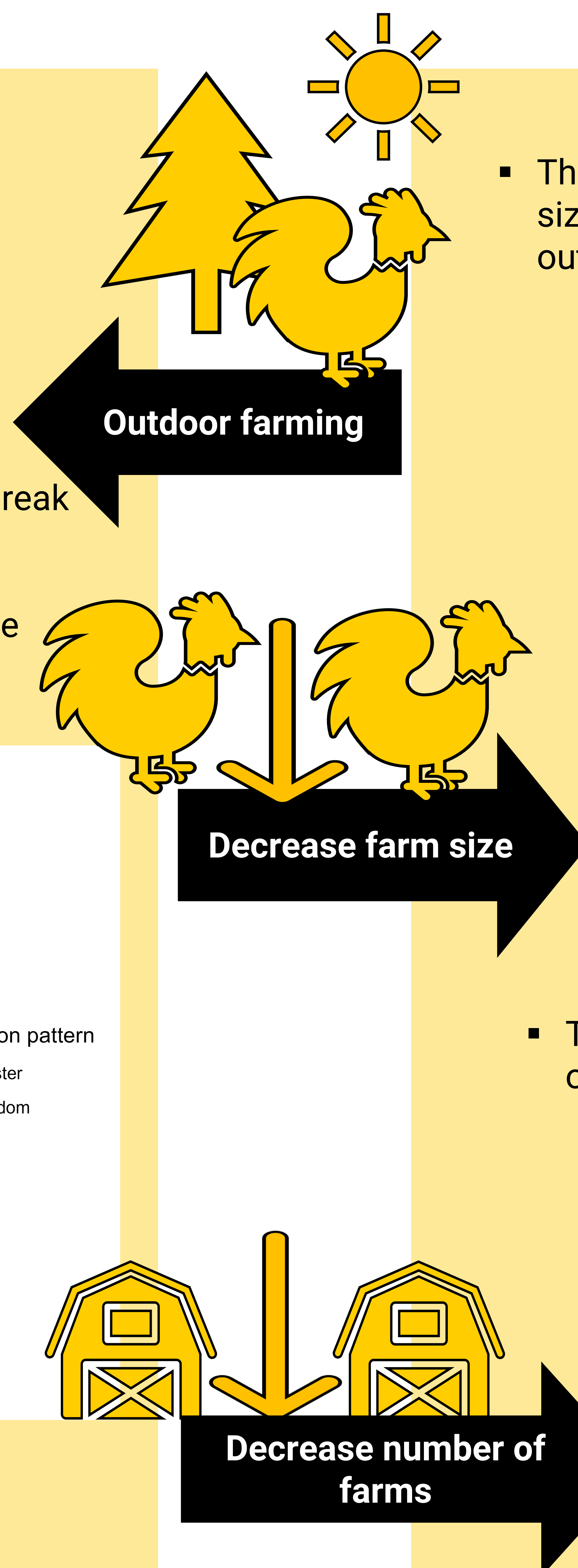
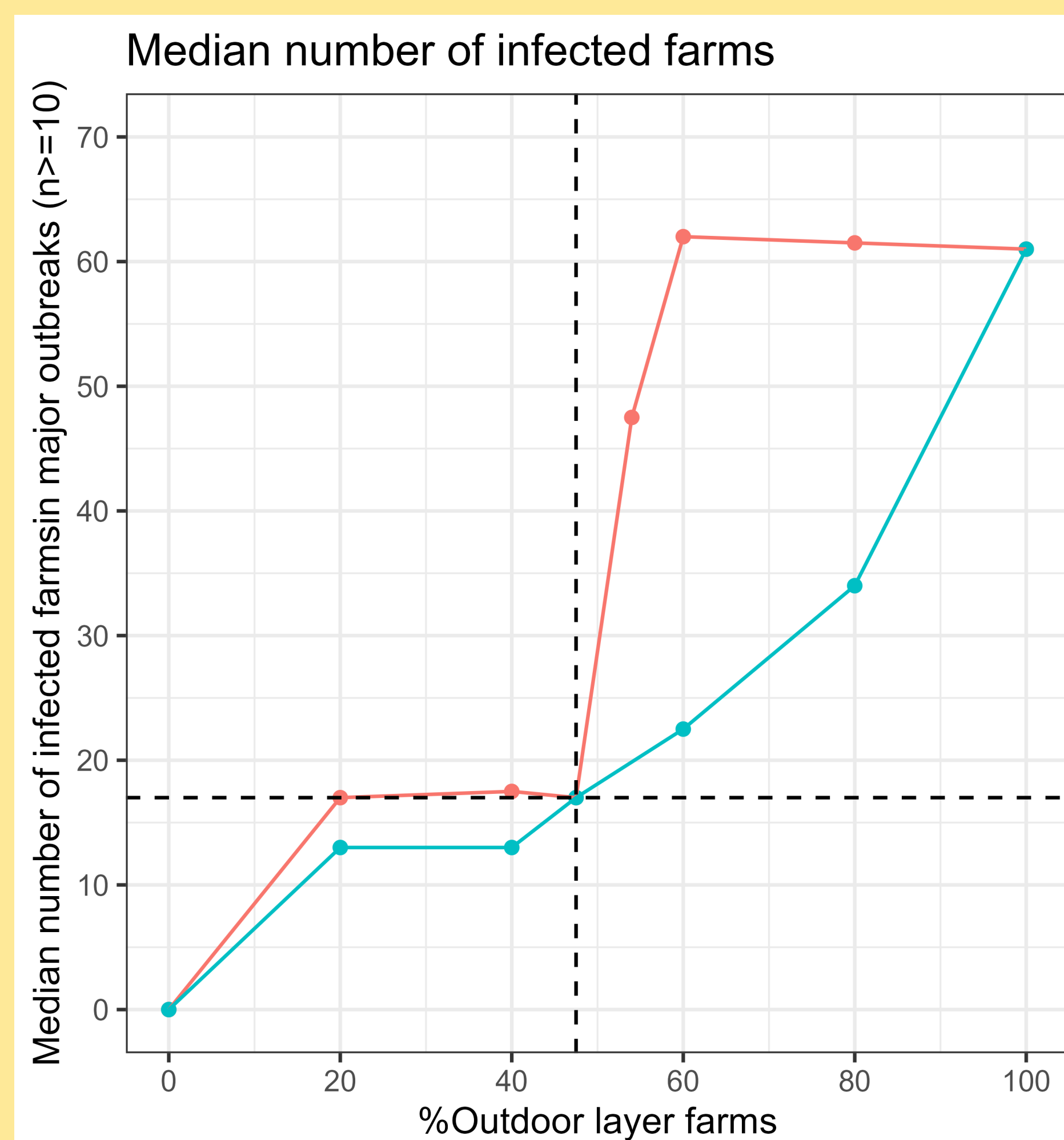
Under different scenarios relative to the current situation:

- Calculate the expected risk of HPAI introduction from environmental sources depending on farm type
- Simulate size and duration of HPAI outbreaks with between-farms spatially explicit transmission model
- Determine the change in human exposure

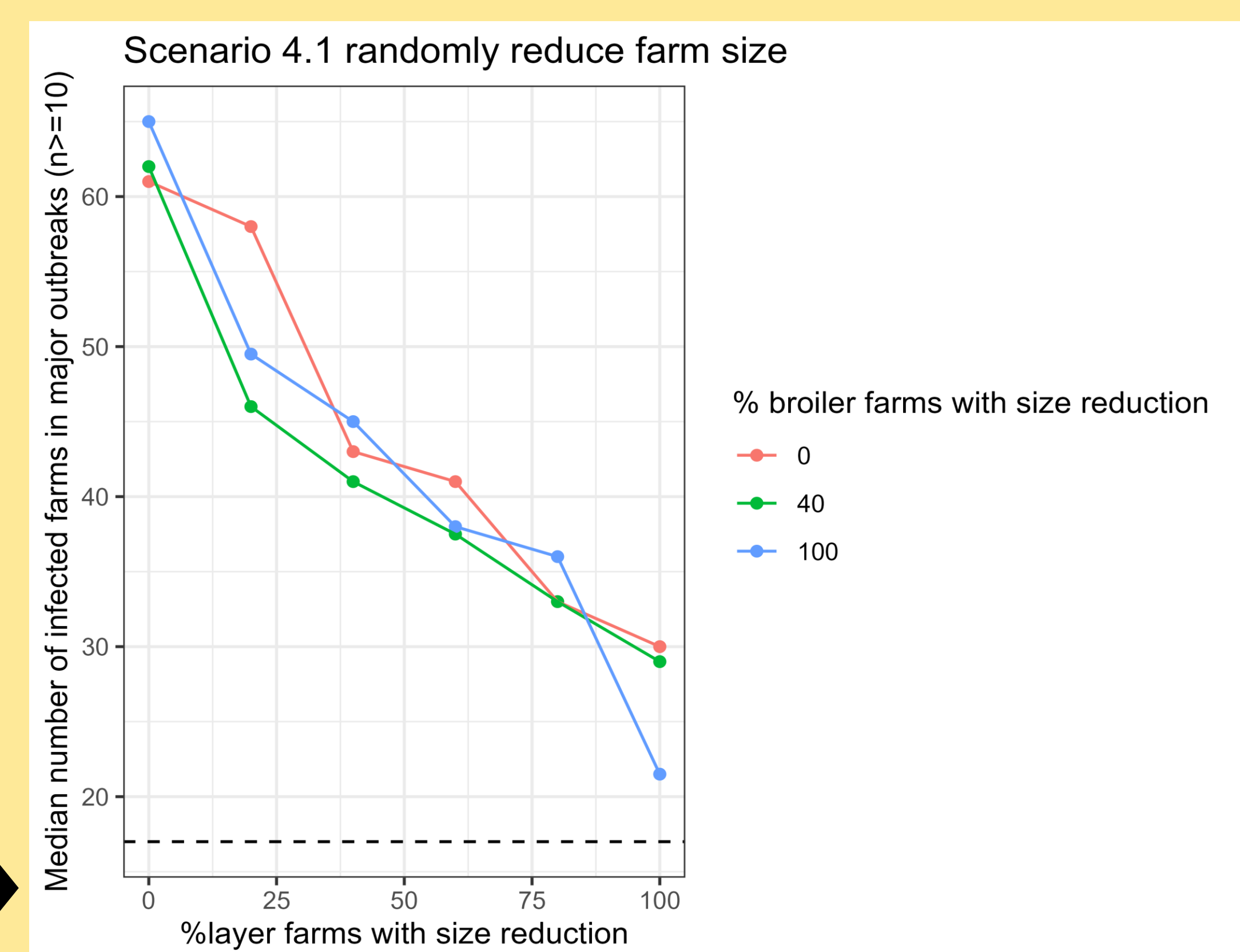
3) Results

Outdoor farming

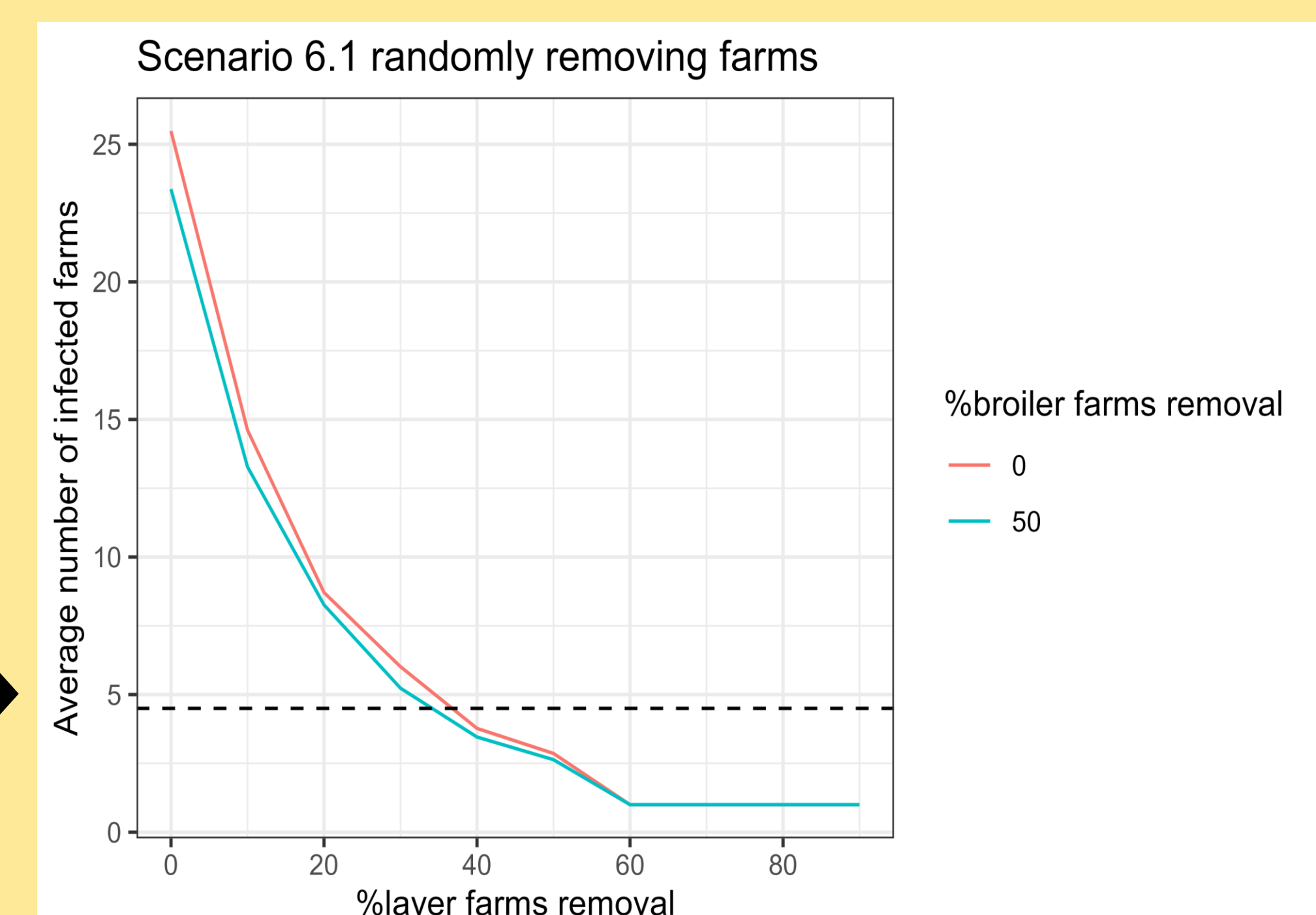
- Increase of HPAI introduction, but can be mitigated by location further from water sources and grassland
- Increased number of infected farms and duration of outbreak.
- Clustering of outdoor farms increases outbreak size
- Increased human exposure compared to the baseline.



- The farm size reduction reduces the outbreak size but will not fully neutralize the effect of outdoor farming.



- The decrease in number of farms reduces the outbreak size. Potential to below baseline



4) Key message:

- The transition to outdoor farming will increase the risk of outbreaks of zoonotic diseases, therefore this transition should proceed cautiously to ensure a balanced approach that integrates animal health, human health, and environmental health.